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2 May 1969

MILITARY SPECIFICATION

# DETONATOR, STAB, M98, (T83E1) PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification covers the inspection of the parts and the loading assembling and packing of a stab type detonator designated as Detonator, Stab, M98.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-398 - RDX
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives; Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9243921 - Detonator, Stab, M98
9258182 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

9243921 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The detonator assembly shall comply with all requirements specified on Drawing (dwg.) 9243921 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Primer mix NOL #130.-The moisture content at the loading station at the time of loading the detonators shall not exceed 0.30 percent when determined as specified in 4.4.1.1.

3.3.2 Lead azide.-The moisture content at the loading station at the time of loading the detonators shall not exceed 0.50 percent when determined as specified in 4.4.1.2.

3.3.3 Cyclonite (RDX).-The moisture content at the loading station at the time of loading the detonators shall not exceed 0.50 percent when determined as specified in 4.4.1.3.



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3.4 Functioning.-The detonator assembly shall not fail to fire on the first attempt and shall comply with the following requirements.

3.4.1 Output.-The detonator assembly shall function under an applied energy of 6 inch ounces maximum (max.) and shall cause a 7/32 inch minimum (min.) diameter perforation in a .131 inch min. thick lead disc when tested, as specified in 4.4.2.1.

3.4.2 Waterproofness.-The detonator assembly shall comply with the requirements of 3.4.1 when conditioned and tested as specified in 4.4.2.2.

3.5 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.6 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the



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Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawings and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings on the moisture content of each explosive shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 complete sets of parts and subassemblies (i.e., 50 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be



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submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Components and subassemblies will be subjected by the Government to any of all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or subassembly in the sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with the applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure



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which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

a. Parts from not more than one supplier

b. Primer mix NOL #130 from not more than one batch or a blend of batches which results in homogeneity of all primer mix used in loading a lot.

c. RDX from not more than one batch or a blend of batches which results in homogeneity of all RDX used in loading a lot.

d. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and the AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, detonator (see dwg. 9243920 covering a detail of dwg. 9243921).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 1.00 percent | | |
| 101. | Length | Gage | 01001 |
| 102. | Outside diameter, min | Gage | 01002 |



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| | | | |
|---|---|---|---|
| 103. | Inside diameter, max | Gage | 01003 |
| 104. | Diameter of hole | Gage | 01004 |
| 105. | Concentricity of hole with outside diameter | Gage | 01005 |
| 106. | Finish on inside of cup improper | Visual | 01006 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) | Visual | 01007 |

4.3.2.2 Disc, Detonator (see dwg. 9243918 covering a detail of dwg. 9243921).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Thickness | Gage | 02001 |
| 102. | Diameter, min | Gage | 02002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) | Visual | 02003 |

4.3.2.3 Disc, detonator closing (see dwg. 9243919 covering a detail of dwg. 9243921).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Thickness | Gage | 03001 |
| 102. | Diameter, min | Gage | 03002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) | Visual | 03003 |

4.3.2.4 Assembly (see dwg. 9243921)

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Cup split, cut or cracked | Visual | 04001 |
| 2. | Explosive on exterior | Visual | 04002 |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.65 percent | | |
| 101. | Total length | Gage | 04003 |
| 102. | Outside diameter, max | Gage | 04004 |
| 103. | Lacquer seal missing or color incorrect | Visual | 04005 |
| 104. | Assembly distorted or damaged to extent that function may be impaired | Visual | 04006 |
| Minor: | AQL 1.50 percent | | |
| 201. | Crimp not full 360 degrees | Visual | 04007 |
| 202. | Disc folded or wrinkled | Visual | 04008 |
| 203. | Evidence of poor workmanship other than critical above (see 3.5) | Visual | 04009 |

4.3.2.5 Box, set-up, after loading and prior to closing slide cover (see dwg. 9258182).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Filler and cushion assembly missing | Visual | 05001 |
| 2. | Filler and cushion assembly improperly assembled | Visual | 05002 |
| 3. | Stiffner missing | Visual | 05003 |
| 4. | Slide cover assembled reversed | Visual | 05004 |
| Major: | AQL 0.40 percent | | |
| 101. | Release paper missing | Visual | 05005 |
| Minor: | AQL 1.00 percent | | |
| 201. | Marking missing, incomplete or illegible | Visual | 05006 |
| 202. | Evidence of poor workmanship (see 3.5) | Visual | 05007 |

4.3.2.6 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top (see dwg. 9258183).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Red surfaces on inner set-up boxes do not alternate on same side of outer set-up boxes | Visual | 06001 |



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2. Stiffner not facing one direction...Visual 06002

Major: None defined

Minor: AQL 1.00 percent

201. Marking missing, incomplete or illegible.....Visual 06003

202. Evidence of poor workmanship (see 3.5).....Visual 06004

4.3.2.7 Box, packing, ammunition prior to assembling cover and strapping (see dwg. 9258184).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Bags containing set-up boxes damaged | Visual | 07001 |
| 102. | Cushioning material incomplete or missing from around outer set-up boxes | Visual | 07002 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) | Visual | 07003 |

4.3.2.8 Box, packing, ammunition (see dwg. 9258184).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Strapping missing, broken, loose or improperly fastened | Visual-Manual | 08001 |
| 102. | Box damaged | Visual | 08002 |
| 103. | Marking missing, broken | Visual | 08003 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) | Visual | 08004 |

4.3.3 Testing



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4.3.3.1 Moisture content (see Table I) - Major defect

TABLE I

| Material | Code No. |
|---|---|
| Primer mix NOL #130 (see 3.3.1) | 09001 |
| Lead azide (see 3.3.2) | 09002 |
| RDX (see 3.3.3) | 09003 |

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one sample of each material from each eight hours production of detonators. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all detonators loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3 as applicable.

4.3.3.2.1 Output (see 3.4.1) - Major defect - Code No. 04010.-The sample size and acceptance or rejection of the lot shall be in accordance with Table II. The test shall be performed as specified in 4.4.2.1 using equipment as specified in 4.3.4.

TABLE II

| Lot size | Sample | Accept | Reject |
|---|---|---|---|
| 1-150,000 | First 315 | 0 | 3 |
| | Second 315 | 3 | 4 |
| 150,001-500,000 | First 500 | 1 | 4 |
| | Second 500 | 4 | 5 |
| 300,001-500,000 | First 800 | 2 | 5 |
| | Second 800 | 6 | 7 |
| 500,001 and over | First 1250 | 3 | 7 |
| | Second 1250 | 8 | 9 |

4.3.3.2.2 Waterproofness (see 3.4.2) - Major defect - Code No. 04011.-Sampling shall be in accordance with MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.2.2.

4.3.3.3 Composition of primer mixture NOL #130 (see dwg. 9243921 - Major defect - Code No. 04012.-The composition of each batch or blend of batches of primer mixture shall be determined at the time of loading. A sample of sufficient size to perform the test specified in 4.4.3 shall be selected.



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If the composition of the sample fails to comply with the requirements, the quantity of primer mixture represented by the sample shall be rejected. Any detonators loaded with the non-conforming mixture shall also be rejected.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET - 9243921 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Primer mixture NOL #130.-Transfer a known weight of between 0.4 to 0.7 gram of the sample to a tared weighing dish approximately 55 millimeters (mm.) in diameter and 30 mm. in height. Place the dish and contents in an oven maintained at 55 to 65 degrees Centigrade (C.) or in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the less in weight as the percentage of moisture.

4.4.1.2 Lead azide.-The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.1.3 RDX.-The moisture content of the RDX shall be determined in accordance with the Method in MIL-R-398.

4.4.2 Functioning

4.4.2.1 Output.-The detonator shall be assembled in the applicable test fixture specified in 4.3.4 so that the firing pin will strike the sensitive end of the detonator. A steel ball weighing two (2) ounces max. shall be dropped from a height of three (3) inches max. on the top of the firing



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pin. The drop height shall be measured as the distance between the top of the firing pin and the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.4.2.2 Waterproofness.-The detonator assemblies shall be conditioned for forty-eight (48) hours min. at a two (2) to three (3) inch depth, in water maintained at seventy (70) plus or minus two (2) degrees Fahrenheit (F.). Immediately after conditioning, the detonator shall be tested as specified in 4.4.2.1. The functioning ability and hole size shall be observed to determine compliance with the requirements.

4.4.3 Composition of primer mixture NOL #130.-The composition of the primer mixture shall be determined as follows:

4.4.3.1 Barium nitrate.-An accurately weighed portion of approximately 0.5 grams of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml. capacity. Three ml. of lead azide saturated distilled water shall be added at a temperature of 50 degrees C. plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six. 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{M(A-B)}{1-.01M}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.4.1.1 in percent.



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4.4.3.2.1 Basic lead styphnate (when applicable).-The residue obtained in the determination of barium nitrate shall be extracted with six-ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this selection shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm.) and its absorbance density determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\,(A-B)}{W(1.01M)D}$$

where:

A = optical density of basic lead styphnate solution
B = optical density of ammonium acetate solution
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.4.1.1 in percent
W = weight of sample
F = factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.



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4.4.3.2.2 Determination of normal lead styphnate (when applicable).-Extract the residue obtained in the barium nitrate determination with sufficient 5 ml. portions of ammonium acetate solution (saturated) at about 25 degrees C. until the washings are colorless. Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml. calibrated volumetric flask, washing th beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml. aliquot of this solution to a second 100 ml. calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of approximately 1 cm. and determine its absorbance density at a wave length of 410 millinicrons. Use as a reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millinicrons, the absorbance density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal lead styphnate, percent} = \frac{F(A-B)}{WD}$$

where:

A = optical density of normal lead styphnate solution.
B = optical density of ammonium acetate solution.
D = width of corex cell, cm.
W = weight of the dried sample.
F = factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

4.4.3.3 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C.



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plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.
W = weight of the sample.
M = moisture content of sample obtained from 4.4.1.1 in percent.

4.4.3.4 Antimony sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1.01M)}$$

where:

D = weight of crucible and sample after testing with hot water.
E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.4.1.1 in percent.

4.4.3.5 Lead azide.-The lead azide content in the sample on a dry basus, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Packaging shall be as specified on dwgs. 9258182 and 9258183.

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5.2 Packing

5.2.1 Level C.-Packing shall be as specified on dwg. 9258184.

5.3 Marking.-Marking of all packaging and packing shall comply with dwgs. 9258182, 9258183 and 9258184.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.



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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army design.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsiblility to the contractor. Contractor designs shall be supported by detailed drawings, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the



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appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Check test for possible deterioration.-The following test is designated to prevent loading of detonators which have detoriorated into the ammunition component with which it is to be loaded. The tests should be performed on detonators from each lot which have been stored for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded not require that the check tests be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.4.1 Functioning.-A sample of 315 detonators shall be selected from each lot and tested as specified in 4.4.2.1. If 3 or more detonators fail to comply with the requirements, the lot shall be rejected. If one or two defectives are found, a second sample of 315 detonators shall be selected and tested. If the total number of defectives from both samples equals or exceeds five, the lot shall be rejected.

6.5 Submission of test data and ammunition data cards.-In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-QA-A-M
(2) SMUPA-ADEP-2

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-001



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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